#	Pillar	Title	Authors	Year	URL	Problem being addressed	Method / approach	Dataset or setting	Evaluation metrics	Main findings	Limitations mentioned by the authors	What assumptions the method makes	What the paper does not test
1	1	On Calibration of Modern Neural Networks	Guo et al.	2017	https://arxiv.org/abs/1706.04599	Modern deep neural network classifiers give confidence scores that don't match their actual accuracy (miscalibration).	Post-hoc calibration: learn a simple transformation of the logits on a held-out validation set, without retraining. Compares temperature scaling, Platt scaling, vector/matrix scaling, histogram binning, isotonic regression, and Bayesian Binning into Quantiles (BBQ).	images and text. For the **images**: 1- Caltech-UCSD Birds 2- Stanford Cars 3- ImageNet 2012 4- CIFAR-10 5- CIFAR-100 6- SVHN For the **text**: 1- 20 News 2- Reuters 3- SST - binary label 4- SST - 5 class labels	Expected Calibration Error (ECE), Maximum Calibration Error (MCE), Negative Log-Likelihood (NLL), reliability diagrams, classification error.	Deeper/higher-capacity modern nets are more overconfident than older shallow nets despite being more accurate. Depth, width, low weight decay, and Batch Normalization are associated with worse calibration. Temperature scaling (one learned scalar) beats every more complex method on almost every dataset.	Authors note it is future work to understand why these architectural trends hurt calibration (correlation shown, not mechanism). Temperature scaling fails to help on Reuters, which was already well-calibrated. Matrix scaling overfits badly as class count grows.	Train/val/test are i.i.d. from the same distribution (no distribution shift). Single, well-defined ground-truth label per input (closed-set classification). A held-out validation set is available for the post-hoc fit.	Anything generative or open-ended (no free-text output). No notion of ambiguous inputs with multiple plausible answers. No decision policy built on top of the confidence score. No dialogue, no multi-turn interaction, no speech/audio.
2	1	Language Models (Mostly) Know What They Know	Kadavath et al.	2022	https://arxiv.org/abs/2207.05221	Whether language models can evaluate the correctness of their own outputs, both by judging a proposed answer (self-evaluation) and by predicting in advance whether they'll be able to answer a question at all (self-knowledge).	Three techniques: (1) direct calibration via multiple-choice format probabilities; (2) P(True) - model samples an answer, then judges the probability that answer is correct, optionally shown several of its own sampled answers first ('brainstorming'); (3) P(IK) - a finetuned classification head predicting, before answering, the probability of getting the question right, trained on empirical fraction of correct samples at temperature 1.	1- Multiple-choice benchmarks (BIG-Bench, MMLU, TruthfulQA, LogicQA, and QuALITY) 2- open-ended generation tasks (TriviaQA, Lambada, GSM8K, Codex HumanEval, GSM8K, Basic arithmetic problems,...)	ECE / RMS calibration error, AUROC (discriminating correct vs incorrect), Brier score, conditional accuracy on samples labeled 'True'.	Larger models are well-calibrated on multiple-choice when format is right, improving with scale and few-shot prompting. Models can meaningfully self-evaluate via P(True), improving with scale ('verification improves faster than generation'). A trained P(IK) head can predict in advance whether the model will get a question right and partially transfers across task domains, but is poorly calibrated out-of-distribution. P(IK) rises appropriately when relevant source material or hints are added to context.	RLHF-finetuned policies are less calibrated than raw pretrained models, so methods may not transfer well to instruction-tuned/chat models. The work tests self-knowledge only as agreement with 'what humans say', not objective truth. Doesn't address deceptive behavior. Generalization results are based on only five datasets, described as 'limited in power and scope'.	Each question has one correct answer to verify against. Single-turn Q&A - no notion of conversation or asking the user something back. The model's own P(True)/P(IK) judgment is assumed a meaningful, stable signal. Text-only input, no acoustic/transcription noise.	No clarification behavior - model can only answer or express low confidence, never ask a follow-up question. No ambiguous queries - every question assumed well-specified. No voice/speech, no noisy input. No RLHF/chat-tuned models in main experiments (flagged by authors). No downstream action tied to the confidence score.
3	1	Teaching Models to Express Their Uncertainty in Words	Lin et al.	2022	https://arxiv.org/abs/2205.14334	Standard calibration methods rely on model logits/token probabilities, which reflect uncertainty over how a claim is phrased, not whether the claim is true (epistemic uncertainty). Asks whether a model can directly state confidence in natural language and have that statement be calibrated.	Finetune GPT-3 on a training set pairing a question, the model's own zero-shot answer, and a target confidence value, so the model learns to output a 'verbalized probability' as language rather than reading it off logits. Compared against 'answer logit' (normalized log-probability of the model's own answer) and 'indirect logit' (log-probability assigned to the token 'True' when asked to judge its own answer).	CalibratedMath: a self-built suite of 21 programmatically generated elementary arithmetic tasks	Mean Squared Error (MSE) between stated confidence and whether the answer was actually correct; Mean Absolute Deviation (MAD). Calibration metrics, not accuracy metrics - answer held fixed, only confidence statement is scored.	Verbalized probability is well-calibrated in-distribution and remains moderately calibrated under distribution shift. Not just parroting logits (outperforms both logit baselines) and not just superficial heuristics (a logistic-regression model using known difficulty cues does worse). Evidence suggests GPT-3 draws on latent pre-trained representations correlated with its own epistemic uncertainty.	The distribution shift tested only varies label distribution and mathematical content - question format stays constant (always short arithmetic Q&A); never tested shifting to chat, long-form QA, or other subject domains like history/biology. Only one base model (GPT-3) tested. Only supervised finetuning used; authors suggest RL as a valuable alternative to try.	Requires a labeled training set of (question, answer, correct/incorrect) to finetune the confidence-expression behavior - a supervised, trained capability. The model's own answer is fixed/held constant. Each question has one or more objectively checkable correct answers (even the 'multi-answer' set is still closed-form and checkable).	No dialogue or interaction - no way for the model to ask a clarifying question if uncertain. No voice/speech, no ASR noise. No natural, open-ended text (everything is short deterministic math). No linkage between calibrated confidence and any downstream action (abstaining, deferring, asking for more information) - abstention is mentioned only in Related Work as other people's work.
4	1	Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback	Tian et al.	2023	https://arxiv.org/abs/2305.14975	RLHF-finetuned LLMs (ChatGPT, GPT-4, Claude, Llama-2-Chat - the models actually deployed) are suspected to have much worse-calibrated log-probabilities than raw pretrained models, since RLHF pushes probability mass toward the preferred answer rather than matching true answer frequency.	Purely prompting-based, no finetuning. Compares: Label prob. (conditional probability from repeated sampling); 'Is True' prob. (Kadavath-style self-evaluation via sampling); Verbalized 1-stage (model gives k answer guesses + numeric confidence in one response); Verbalized 2-stage (answer first, then a separate turn asks for probability); Verbalized + chain-of-thought; Linguistic expressions ('Likely', 'Almost certain') mapped to probabilities via human survey or 'optimized' against a held-out calibration set.	3 QA datasets: TriviaQA, SciQ, TruthfulQA they were tested across 5 RLHF-tuned models: **gpt-3.5-turbo**, **gpt-4**, **claude-1**, **claude-2**, **Llama-2-70b-chat**	ECE and temperature-scaled ECE, Brier Score (temperature-scaled), AUC (selective classification: accuracy vs coverage curve).	Verbalized confidence (numeric or linguistic) is consistently better calibrated than the model's own conditional probability across GPT-3.5, GPT-4 and Claude, often reducing ECE by ~50%, no finetuning required. Generating multiple candidate answers before stating confidence improves calibration. Chain-of-thought prompting does NOT help verbalized calibration. Effect is inconsistent for Llama-2-70B-Chat - open-source model benefits far less than closed proprietary ones, and the paper cannot explain why since architectural/training details of closed models are unavailable.	Authors explicitly state: (1) restricted to factual-recall QA - unclear if results hold for reasoning-heavy tasks; (2) lack of transparency about closed RLHF-LMs' training limits ability to explain why calibration differs across models; (3) restricted to short-form QA - long-form generation explicitly flagged as future work.	Same closed-form, single-correct-answer assumption as prior calibration work (even with LLM-judge semantic equivalence, one 'correct' target per question). Assumes the confidence-elicitation prompt itself doesn't change the answer. Assumes access to enough API budget for repeated sampling (n=10) per question.	Long-form or open-ended generation (explicitly flagged as future work). No dialogue/multi-turn interaction beyond mechanical 2-stage prompting - no clarifying questions to the user. No ambiguity in the questions themselves. No voice/speech/ASR layer. No decision policy - calibration measured and reported, but nothing done with the confidence score (no abstention, no deferral, no clarification triggered by low confidence).
5	1	Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs	Xiong et al.	2024	https://arxiv.org/abs/2306.13063	Prior confidence-elicitation methods rely on white-box access to internal model information or fine-tuning, which is becoming unsuitable for closed-source commercial LLM APIs. Explores black-box approaches for LLM uncertainty estimation.	Defines a systematic framework with three components: prompting strategies (Vanilla, Multi-step, Self-Probing, Top-K, CoT) for eliciting verbalized confidence; sampling strategies for generating multiple responses; aggregation techniques (e.g. Avg-Conf, Pair-Rank, Consistency) for computing confidence from response consistency. Combines components into different algorithms.	**Five types of tasks:** 1- Commonsense 2- Arithmetic 3- Symbolic 4- Ethics 5- Professional Knowledge across eight datasets, evaluated on five widely-used LLMs: GPT-3, GPT-3.5, GPT-4, Vicuna, LLaMA 2	ECE, AUROC, failure prediction performance across the two benchmarked tasks (confidence calibration and failure prediction).	LLMs tend to be highly overconfident when verbalizing confidence, with values predominantly falling in 80-100% range in multiples of 5 (similar to how humans talk about confidence); scaling model capacity improves but doesn't fix this. Human-inspired prompting strategies mitigate overconfidence but improvement diminishes as model capacity scales. Sampling and aggregation strategies improve failure prediction. White-box methods perform better than black-box but the gap is narrow (0.522 to 0.605 in AUROC). Methods still struggle on tasks requiring specialized/professional knowledge	Explicitly stated 'Limitations and Future Work': (1) Scope of Datasets - focuses mainly on fixed-form and free-form QA tasks with a unique ground truth answer, leaving summarization and open-ended QA to future work; (2) Black-box Setting - black-box approaches remain suboptimal versus white-box; integrating black-box methods with limited white-box access (e.g. GPT-3 logits) suggested as a promising direction.	Assumes each question has a unique ground-truth answer to check confidence against. Assumes black-box API access (no logits) is the deployment reality worth optimizing for. Assumes repeated sampling (multiple responses per question) is computationally feasible.	Summarization and open-ended QA (explicitly future work). No dialogue/clarification behavior - only answer + confidence, no way to ask the user anything. No voice/speech/ASR. No decision policy connecting confidence score to an action like abstention or clarification.
6	1	Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation	Kuhn, Gal & Farquhar	2023	https://arxiv.org/abs/2302.09664	Measuring uncertainty in free-form natural language generation is challenging because of 'semantic equivalence' - different sentences (different token sequences) can mean the same thing, so token-level uncertainty doesn't capture uncertainty over meaning.	Introduces semantic entropy: an algorithm that clusters sampled model outputs into groups that mean the same thing (via bidirectional entailment using a smaller NLI model), then computes the entropy of the probability distribution over these meaning-clusters rather than over raw token sequences. Unsupervised, single model, no finetuning or architectural modification required.	Free-form question answering: TriviaQA and CoQA (open- and closed-book). Uses OPT models of varying size (2.7B to 30B parameters).	AUROC (predicting model accuracy from the uncertainty measure). Compared against baselines: predictive entropy, normalised entropy, lexical similarity (Rouge-L), and P(True)/entropy reimplementations of Kadavath et al.	Semantic entropy predicts model accuracy on free-form QA better than all baselines and this outperformance increases with model size while also doing well for smaller models. Makes better use of additional samples than baselines because it handles duplicate/paraphrased samples correctly. An intermediate sampling temperature (0.5) balances the diversity/accuracy trade-off best for uncertainty estimation, contrasting with prior work's use of temperature 1.0.	Explicitly noted in Discussion/future work: distribution-shift tests only varied label distribution and content, not question format (always short arithmetic-style QA in comparable prior work) - specifically for this paper, the method's paraphrase-identification step is only tested for QA with relatively clear ground truths; extending to summarisation (which requires more human evaluation) needs further progress on paraphrase identification first. Method does not protect against a model capable of deception (only measures uncertainty between meaningful options).	Assumes questions have relatively clear ground truths (chosen specifically because this is 'easier to establish... than more nebulous tasks like summarisation'). Assumes bidirectional entailment (inference in both directions) is a valid operationalization of 'meaning the same thing'. Requires access to sample multiple generations from the model (compute cost).	Summarisation or other nebulous/long-form tasks (explicitly future work). No dialogue, no clarification behavior - the method quantifies uncertainty but does not act on it. No voice/speech/ASR uncertainty. No ambiguous user intent - the ambiguity modeled is about which token sequences mean the same thing, not about what the user wants.
7	1	A Survey of Confidence Estimation and Calibration in Large Language Models	Geng et al.	2024	https://arxiv.org/abs/2311.08298	No comprehensive survey exists organizing the growing body of research on LLM confidence estimation and calibration, despite it being crucial for mitigating hallucination/unreliability risk.	Survey/taxonomy paper - no original experiments. Organizes literature into: confidence/uncertainty/calibration concepts and metrics (classification vs generation); estimation methods for generation tasks (logit-based, internal-state-based, linguistic confidence, consistency-based, surrogate models) and classification tasks; calibration improvement methods; and applications.	-	-	Identifies that LLMs pose unique challenges versus discriminative models: output space grows exponentially with generation length making exhaustive enumeration impossible; different expressions can convey the same meaning (semantic equivalence) requiring confidence estimation to consider semantics; LLMs can express confidence in words and perform zero/few-shot learning but are prompt-sensitive.	Explicit 'Limitations' section: (1) No experimental benchmarks - as a survey with no original experiments, cannot offer empirical validation of the theories/concepts discussed, limiting its ability to contribute new, verified knowledge; (2) Potential omissions - due to rapid development in the field, some important work may have been overlooked despite best efforts to compile latest advancements.	N/A - as a survey, the paper does not itself make methodological assumptions, but summarizes assumptions of the underlying literature (e.g. that confidence and uncertainty are inversely related; aleatoric/epistemic uncertainty distinction).	No original benchmarking across tasks/domains (explicitly named as needed future work, requiring extensive human annotation especially for long-form generation). Multi-modal LLM calibration - unclear whether existing methods transfer, flagged as open question. Calibration to human variation/disagreement (task ambiguity, annotator subjectivity, input ambiguity) - flagged as a future direction, not addressed by the survey's own analysis.
8	1	Position: Uncertainty Quantification Needs Reassessment for Large-language Model Agents	Kirchhof et al.	2025	https://arxiv.org/abs/2505.22655	Traditional uncertainty quantification decomposes uncertainty into aleatoric (irreducible, from noisy data) and epistemic (reducible, from lack of model knowledge). This position paper argues that dichotomy is too limited for the open, interactive, multi-turn setting LLM agents operate in when talking with users.	Position/argument paper, not empirical. Reviews the literature and finds popular definitions of aleatoric/epistemic uncertainty directly contradict each other (Table 1 in the paper catalogs conflicting 'schools of thought'). Proposes three research directions instead: (1) underspecification uncertainties - for when users don't provide all information at first; (2) interactive learning - asking follow-up questions to reduce uncertainty about context; (3) output uncertainties - using rich language/speech to express uncertainty as more than a number. Also presents a counter-position/alternative-views section.	-	-	Shows that what initially appears as epistemic uncertainty (lack of knowledge) can become aleatoric if additional information fails to reduce ambiguity, and aleatoric uncertainty (e.g. an underspecified question) can become reducible/epistemic by asking clarifying questions - directly contradicting the traditional dichotomy in interactive settings. Argues traditional numeric uncertainty scores are poorly suited to human-facing chat interactions versus richer textual explanations of uncertainty.	The paper's own 'Alternative Views' section presents counter-positions and partially concedes them: (1) aleatoric/epistemic terms still have value for training-phase and active-learning use even if less applicable in interactive chat; (2) interactive learning might be solvable via next-token prediction on large interaction datasets in some scenarios (e.g. standardized customer-service forms), though the authors argue this doesn't eliminate the need for research on honest question-asking; (3) numeric uncertainty is still preferable when an agent talks to another automated system rather than a human.	Assumes multi-turn, human-facing interaction is the relevant deployment context to reassess UQ for (not e.g. batch classification). Assumes that communicating uncertainty via rich language is generally more useful to a human decision-maker than a bare number, citing evidence of 'blind trust' behavior at high numeric certainty (Kapoor et al. 2024).	As a position paper, it proposes but does not empirically validate: methods for detecting underspecification uncertainty, methods for interactive learning/clarification-question generation, or methods for generating rich textual uncertainty explanations. No voice/speech modality discussed. No concrete evaluation metrics for any of the three proposed research directions.
9	2	Selective Question Answering under Domain Shift	Kamath et al.	2020	https://arxiv.org/abs/2006.09462	QA models need to know when to abstain from answering to avoid giving wrong answers, especially since users often ask questions diverging from the model's training data (out-of-domain/OOD), making errors more likely.	Proposes selective question answering under domain shift: a QA model tested on a mixture of in-domain and OOD data must answer as many questions as possible while maintaining high accuracy. Trains a calibrator (random forest classifier over input + softmax-output features) to predict whether the QA model is correct on a given example, using both in-domain data and data from a known OOD distribution (different from the unknown test OOD distribution).	SQuAD as the source/in-domain distribution; five other QA datasets (via MRQA) as OOD distributions, averaged across 20 choices of unknown/known OOD dataset pairs; tested on a uniform mixture of SQuAD and unknown OOD data.	Coverage at 80% accuracy (% of questions answered while maintaining 80% accuracy), AUC of the risk-coverage curve.	MaxProb (abstaining based on the model's own softmax probability) is well-calibrated in-domain but overconfident on OOD data, so it performs poorly in the mixed setting. Training a calibrator using known OOD data achieves 56.1% coverage at 80% accuracy, versus 48.2% for MaxProb alone, 51.8% for training the QA model on source+known-OOD data, and 53.7% for a calibrator trained on source-only data. OOD data - even from a different distribution than the true test OOD - improves selective prediction under domain shift.	No explicit 'Limitations' section (pre-dates that convention), but Discussion notes the framework is general and 'extends to any prediction task' beyond QA - implying the specific numeric results are QA-specific. Appendix notes question-length/word-overlap features did not improve the calibrator; an outlier-detection approach to selective prediction (distinguishing in- vs out-of-domain rather than predicting correctness) performed poorly (AUC 24.23).	Assumes access to a small sample of data from a 'known' OOD distribution, even though the true test-time OOD distribution is unknown - a specific and non-trivial assumption about what the system developer can foresee. Assumes correctness of an extractive QA answer is well-defined (exact match style), though multiple valid spans exist for SQuAD (addressed via evaluation restriction ablation in the Appendix).	No generative/open-ended QA, only extractive span-based QA. No dialogue - abstention is a single binary decision (answer/abstain), with no notion of asking a clarifying question instead of abstaining. No voice/speech/ASR uncertainty. No connection between OOD detection at the linguistic/semantic level and the underlying cause of ambiguity (the paper doesn't distinguish acoustic vs. intent vs. underspecification-type uncertainty, since text-only setting).
10	2	Know Your Limits: A Survey of Abstention in Large Language Models	Wen et al.	2024	https://arxiv.org/abs/2407.18418	LLMs tend to hallucinate, generate overly certain responses, answer with incomplete information, or produce harmful responses; in these situations models should ideally abstain (refuse to answer under uncertainty), but existing work on abstention is fragmented.	Survey paper - no original experiments. Introduces a framework analyzing abstention from three perspectives: query (is the input ambiguous/incomplete/beyond any possible knowledge/lacking context/containing knowledge conflicts?), model knowledge (is the model confident enough in correctness?), and human values (does the response align with human values/avoid harm?). Categorizes existing abstention methods by model lifecycle stage: pretraining, alignment, inference.	-	-	Identifies five major types of abstention expression from prior work: cannot assist; refutes the query; provides multiple perspectives without preference; perceives risk and answers cautiously with disclaimer; refuses due to lack of knowledge/certainty. Identifies over-abstention (unnecessary refusal, e.g. from excessive 'self-check' techniques or over-focused instruction tuning) and vulnerability of abstention to prompt wording/social engineering as key open problems (Section 5).	No dedicated 'Limitations' section identified in the extracted text, but the paper's own Future Directions section functions similarly: calls for abstention as a 'meta-capability' across tasks (currently fragmented by task); more generalizable evaluation; better customization to user needs; abstention-aware privacy/copyright protection; generalizing abstention beyond LLMs to vision/multimodal; improving multilingual abstention given performance gaps between high- and low-resource languages.	Assumes abstention can be meaningfully decomposed into query/model/human-values perspectives that are 'interconnected but distinct'. Assumes a workflow where a single LLM f produces one output y per input x, with abstention as a binary or partial modification of that one output (not as part of a longer clarifying dialogue).	As a survey, does not itself empirically test anything. Notably, its abstention framework treats abstention as a terminal act ('refuse to answer') rather than as part of an interactive clarification exchange - asking a clarifying question is not modeled as a distinct category from abstention. No voice/speech dimension. No discussion of over-abstention/vulnerability specifically in the context of ambiguous vs. simply-hard queries.
11	3	AmbigQA: Answering Ambiguous Open-domain Questions	Min et al.	2020	https://arxiv.org/abs/2004.10645	Open-domain QA systems assume questions have a single well-defined answer, but ambiguity arises frequently in real search-style queries (over 50% of a sampled set of Google search queries were found ambiguous).	Introduces the AmbigQA task: find every plausible answer to a question, then rewrite the question for each one to resolve the ambiguity (rather than asking the user a clarifying question). Extends a state-of-the-art QA model (DPR/Karpukhin et al.) with three components: set-based QA with a sequence-to-sequence model, a question disambiguation model, and a modified democratic co-training approach leveraging partial supervision from the full NQ-OPEN dataset.	AMBIGNQ: 14,042 questions from NQ-OPEN (an open-domain version of Natural Questions), annotated by having annotators search, navigate, and read multiple Wikipedia pages to find as many distinct answers as possible. 2.1 distinct answers per question on average, 89.0 F1 inter-annotator agreement.	F1ans (multi) for zero-shot multi-answer prediction standard QA accuracy metrics on unambiguous subsets.	Diverse ambiguity types identified via 100-sample breakdown: event references (39%), properties (27%), entity references (23%), answer types (16%), time-dependency (13%), multiple sub-questions (3%) - many overlapping. Baseline models rarely get all correct answers when there are multiple references, often generating only a subset (under-generation). Accuracy on examples with a single answer is high (13/20 in error analysis), suggesting NQ-OPEN substantially underestimates true model performance due to prevalence of unmarked ambiguity.	Explicit 'Conclusion & Future Work': recall of multiple answers, not disambiguated-question generation, is identified as the primary remaining challenge. Future work suggested: explicitly modeling ambiguity over events/entities or in the retrieval step; applying the approach to structured-data QA (tables); handling no-answer or ill-formed questions with more complex ambiguous information needs; more carefully evaluating usefulness to end users.	Assumes ambiguity can be fully resolved by providing multiple answers + disambiguated question rewrites, rather than by interacting with the user - i.e., assumes a non-interactive, single-shot answering paradigm is sufficient. Assumes annotators (and by extension, models) can enumerate 'every plausible answer', though the paper itself notes this is 'inherently difficult to know' even for human experts.	No interactive clarification (the system never asks the user a question - it always tries to answer all interpretations at once). No voice/speech modality. No decision policy for when disambiguation is worth the extra output length versus giving a single best answer. No evaluation of end-user usefulness of the multi-answer format (explicitly flagged as future work).
12	3	ConvAI3: Generating Clarifying Questions for Open-Domain Dialogue Systems (ClariQ)	Aliannejadi et a;.	2020	https://arxiv.org/abs/2009.11352	Some user requests in dialogue systems are ambiguous (having more than one possible answer); unlike IR settings, where this is handled via search-result diversification, dialogue settings require the system to identify ambiguity and ask a good clarifying question instead of answering directly.	This is a shared-task/challenge description document, not an experimental paper. Defines two research questions - RQ1: when to ask clarifying questions; RQ2: how to generate them - and a two-stage challenge design: Stage 1 uses a static crowdsourced dataset (user request + necessity-of-clarification label 1-4 + candidate clarifying questions + user answers) for initial training/testing via retrieval/ranking; Stage 2 puts participating systems in front of real human users in a human-in-the-loop setting requiring a generative approach.	Crowdsourced dataset (inspired by the Qulac dataset, itself based on TREC Web Track 2009-2012): 237 topics, 891 facets, 3,304 questions, 11,489 question-answer pairs, split 70% train / 30% validation. Question bank provided for Stage 1 testing.	Stage 1 (document retrieval effect of clarification): MRR, Precision@[1,3,5,10,20], nDCG@[1,3,5,20]; question-relevance ranking: Recall@[10,20,30]. Stage 2 (human-in-the-loop): user ratings of Relevance and Naturalness per system turn, plus overall interaction rating.	N/A - this is a challenge/task design document, not a paper reporting experimental results; it defines the setup and metrics that participating systems would later be evaluated on.	N/A - no explicit limitations section, as this is a task description rather than a study with its own experimental results to critique.	Assumes clarification necessity can be captured as a 1-4 ordinal label per user request. Assumes a fixed 'question bank' can adequately cover the space of useful clarifying questions for Stage 1's retrieval framing. Distinguishes Relevance from Naturalness as separate evaluation axes on the assumption these are meaningfully different (illustrated by an example of a relevant-but-unnatural 'yes/no puzzle' style clarification dialogue).	As a task/challenge design paper, it does not itself report system performance, error analysis, or generalization results - those would come from later systems submitted to the challenge. No discussion of confidence/calibration signals driving the clarify decision. No voice/speech modality (text-based conversational IR).

13	3	CLAM: Selective Clarification for Ambiguous Questions with Generative Language Models	Kuhn, Gal & Farquhar	2022	https://arxiv.org/abs/2212.07769	Current LLMs rarely ask users to clarify ambiguous questions, instead providing incorrect answers based on one of several possible interpretations - the fact that user requests are often ambiguous is well-established in IR but had received little attention in the LLM QA community at time of writing.	Introduces CLAM: a framework prompting LLMs to (1) detect via few-shot classification whether a given question is ambiguous, (2) generate an appropriate clarifying question, and (3) give a final answer after receiving clarification. Frames this as a form of 'meta-cognition' (thinking about thinking). Also introduces an automatic evaluation protocol simulating users via an 'oracle' LM given privileged information, to allow scalable multi-turn dialogue evaluation without human subjects.	Introduces '**Ambiguous TriviaQA**': 200 pairs of ambiguous/unambiguous questions derived from **TriviaQA** by replacing a name/noun with a generic pronoun or similar transformations, such that exactly one piece of clarifying information disambiguates each pair. Also evaluates on **CLAQUA I**, **CLAQUA II**, and **ClariQ** (existing datasets)	**Accuracy** (correct initial answer vs. correct answer after clarification, fraction of correct clarifications provided by the oracle model.	Prompting the model to ask for clarification significantly improves QA accuracy on ambiguous questions across Ambiguous TriviaQA, CLAQUA I, and CLAQUA II, without affecting performance on unambiguous questions. Few-shot prompting is a highly reliable way to detect whether a question is ambiguous, avoiding unnecessary clarification questions on precise inputs. The oracle model reliably provides correct clarifying information on TriviaQA and CLAQUA II; on CLAQUA I (disambiguating two meanings of a proper noun) it sometimes gives information applying to both entities, failing to disambiguate	No dedicated 'Limitations' section identified in the extracted text. In Related Work, the paper contrasts itself with a concurrent fine-tuning approach (Krasheninnikov et al. 2022) that requires 175B-parameter finetuning and still has a 44.5% false omission rate for recognizing ambiguous inputs, noting CLAM's approach requires no finetuning either for the QA model or the oracle.	Assumes ambiguity can be reduced to needing exactly 'one piece of clarifying information' per question (baked into the Ambiguous TriviaQA construction). Assumes an oracle LM with privileged information can validly stand in for a real human user during automatic multi-turn evaluation. Assumes few-shot prompting alone (no finetuning) is sufficient for reliable ambiguity detection.	No real human-user evaluation (only oracle-LM simulated users) - explicitly framed as a scalability trade-off since human multi-turn evaluation is 'expensive, hard to reproduce, and can require ethics-board approval'. No voice/speech modality. No acoustic/mishearing-type ambiguity - only text-based semantic/entity ambiguity. No exploration of how many clarification turns are needed when more than one piece of information is missing.
14	3	Clarify When Necessary: Resolving Ambiguity Through Interaction with LMs	Zhang & Choi	2023	https://arxiv.org/abs/2311.09469	Ambiguity is embedded throughout language; interactive AI assistants (ChatGPT, Claude, LLaMA-2) often fail to interact with users to resolve ambiguity in their requests, despite ambiguity resolution through interaction being a hallmark of natural human language use.	Proposes a task-agnostic framework decomposing ambiguity resolution into three subtasks: (1) determining when clarification is needed, (2) determining what clarifying question to ask, (3) responding accurately using the clarification received. For subtask 1, introduces INTENT-SIM: a novel uncertainty-estimation approach that determines the utility of querying for clarification by estimating the entropy over simulated user intents (multiple simulated user-assistant interactions).	Draws ambiguous examples from three existing NLP datasets/settings, evaluated across three application types: question answering (built on Min et al. 2020's AmbigQA-style data), machine translation (Bawden et al. 2018), and natural language inference (Liu et al. 2023). Uses multiple annotations per example to derive a natural distribution over user intents.	Performance gain over randomly selecting examples to clarify, at a fixed clarification budget (e.g. clarifying on only 10% of examples); comparison against existing uncertainty-estimation baselines for identifying which predictions benefit from clarification.	When only allowed to ask for clarification on 10% of examples, INTENT-SIM doubles the performance gains over randomly selecting examples to clarify. INTENT-SIM consistently outperforms existing uncertainty-estimation approaches at identifying predictions that will benefit from clarification, and is robust across a wide range of NLP tasks and LMs.	No dedicated 'Limitations' section was found in the pages examined. The Conclusion frames the work as 'laying the foundation' for future work, suggesting current scope is a starting point rather than a complete solution: future work is suggested for developing task-agnostic clarifying-question generation methods, extending to multi-turn interactions, and examining what is learned through chat-finetuning.	Assumes access to multiple human annotations per ambiguous example, used to derive the natural distribution over user intents needed for realistic 'when to clarify' evaluation - a data requirement not available for arbitrary real-world queries. For the second/third subtasks, an oracle-generated clarifying QA pair corresponding to the gold interpretation is provided to the system rather than being generated by the system itself, since the paper's own framework figure marks these as inputs from 'existing sources of disambiguated input/output pairs'.	The framework's second subtask (deciding what question to ask) and third subtask (responding using the clarification) are less thoroughly evaluated than the first (deciding whether to ask) - the paper's central contribution (INTENT-SIM) addresses only the 'when' decision. No voice/speech modality. No multi-turn extension (single clarification round only, as noted in the paper's own future-work discussion).
15	3	Tree of Clarifications: Answering Ambiguous Questions with Retrieval-Augmented Large Language Models	Kim et al.	2023	https://arxiv.org/abs/2310.14696	For ambiguous open-domain questions, one approach is to provide a comprehensive long-form answer addressing all interpretations (rather than asking the user to clarify) - but considering multiple dimensions of ambiguity and gathering the corresponding knowledge for each remains a challenge.	Introduces TREE OF CLARIFICATIONS (TOC): recursively constructs a tree of disambiguated questions (DQs) for an ambiguous question (AQ) via few-shot prompting that leverages externally retrieved passages (using ColBERT and Bing search, combined for over 200 passages), pruning unhelpful DQs via self-verification, then aggregates the valid tree nodes into a long-form answer addressing all valid interpretations.	ASQA (Stelmakh et al. 2022), a long-form QA benchmark for ambiguous questions, used in a 5-shot (few-shot prompting) setup rather than full-training-set supervision.	Disambig-F1, Disambig-ROUGE (D-F1, D-ROUGE style metrics measuring whether the long-form answer correctly addresses each disambiguated interpretation), ROUGE-L, Answer-F1 (for intrinsic evaluation of individual disambiguated questions).	TOC outperforms existing baselines on ASQA across all metrics in the few-shot setup, and its 5-shot performance surpasses fully-supervised baselines trained on the whole training set by 7.3 (Disambig-F1) and 2.9 (Disambig-ROUGE) points. Integrating retrieval systems (ColBERT + Bing) substantially improves disambiguation accuracy, especially Disambig-F1. The self-verification pruning method more precisely identifies helpful disambiguations than deduplication alone (+18.4 Answer-F1 over baseline).	Explicit 'Limitations' section: the study is limited in demonstrating generalizability across different kinds/sizes of LLMs (only tested with one setup). Experiments are conducted only on a single benchmark, ASQA. The cost of multiple iterative prompting (recursive tree construction) is non-negligible. Chain-of-thought prompting was tried but failed to improve performance in pilot experiments, suggesting the disambiguation process specifically needs external knowledge rather than just more reasoning steps.	Assumes external retrieval (Wikipedia via ColBERT/Bing) is necessary and sufficient to ground disambiguation in factual correctness, rather than relying on the LLM's own parametric knowledge. Assumes a tree structure (recursive branching) is an appropriate way to represent the space of possible disambiguations, rather than e.g. a flat list.	Generalization across different LLM families/sizes (explicitly flagged as a gap). Generalization beyond the single ASQA benchmark. No interactive clarification with a real user - the system generates a long-form answer covering all interpretations rather than asking the user which interpretation they meant. No voice/speech modality.
16	3	Eliciting Human Preferences with Language Models (GATE)	Li et al.	2023	https://arxiv.org/abs/2310.11589	Selecting examples or writing prompts to specify a task for a language model is challenging, especially for tasks involving unusual edge cases, nebulous preferences, or requiring an accurate mental model of LM behavior - existing methods put the full burden of comprehensive specification on the user.	Introduces generative active task elicitation (GATE): a learning framework where the model itself elicits and infers a user's intended behavior through free-form, open-ended, language-based interaction (e.g. asking informative open-ended questions or generating edge cases for the user to label), rather than the user writing a static prompt or labeling a fixed example pool. Frames existing approaches (supervised learning, active learning, prompting) along two axes: interactivity and flexibility, with GATE occupying the previously-unoccupied 'interactive + free-form' quadrant.	Three domains studied in pre-registered human experiments: email validation, content recommendation, and moral reasoning. Human participants interact with GATE-based elicitation methods versus baseline prompting and active-learning methods; agreement is measured on a held-out test set of the true decisions the user would make.	Alignment between model predictions and the user's true decisions on held-out test cases; a cost/effort measure (e.g. words typed) balanced against alignment, per the paper's formal objective (alpha*cost + beta*alignment).	LM-based task elicitation (GATE) often yields more accurate models than existing prompting or active-learning techniques, while requiring comparable or less mental effort from users, and surfaces novel considerations users had not initially anticipated. Generative elicitation surfaced more novel considerations in moral reasoning and email validation domains than in content recommendation, where edge cases were less tricky/intuitive.	Explicit Appendix F 'Limitations': exploration of GATE methods was limited to prompt-based approaches with no explicit optimization of the stated objective (Equation 2); future work could combine explicit optimization with LM flexibility. Human experiments lacked budget to survey a large number of participants, so statistical significance above baselines could not be established in certain domains; sample may be biased (all English-speaking, US-based). The moral reasoning domain is described as 'very simplistic' and may not capture the full nuance of human moral preference; paper explicitly does not endorse aligning to every possible human preference given ethical risks.	Frames the paper's own contribution as modality-agnostic - explicitly notes GATE 'could be applied to other settings (e.g., speech-based or multimodal models)' though this specific paper studies only language-based elicitation. Assumes appropriate definitions of specification cost (C) and alignment (A) are domain-dependent and must be defined per-application. Assumes users can meaningfully judge and label edge cases generated by the model.	Speech/voice-based elicitation (explicitly named as a modality the framework could extend to, but not tested in this paper). Explicit optimization of the GATE objective beyond prompting (flagged as future work). Statistical significance of GATE's advantage in all domains (limited by sample size, per the authors' own Limitations). Broader population diversity beyond English-speaking US participants.
17	3	STaR-GATE: Teaching Language Models to Ask Clarifying Questions	Andukuri et al.	2024	https://arxiv.org/abs/2403.19154	When prompting language models, users often leave important aspects unsaid (task ambiguity). Asking questions could resolve this, but models often struggle to ask good questions, and prompting closed-source LMs to ask questions (GATE) is inflexible in guiding a model's questioning strategy.	STaR-GATE: combines active preference elicitation (GATE) with a self-improvement loop inspired by STaR (Self-Taught Reasoner). Generates a synthetic dataset of 25,500 unique persona-task prompts simulating conversations between a Questioner (pretrained LM) and a Roleplayer (whose preferences are unknown to the Questioner). The Questioner is iteratively finetuned on questions that increase the log-probability of high-quality 'gold' responses (generated by an Oracle with access to the Roleplayer's latent persona), with response-regularization added to avoid distribution shift.	Synthetic dataset of 25,500 unique persona-task-response prompts, self-generated for the training/simulation loop. Roleplayer during training restricted to mixtral-8x7b-instruct; ablations also test llama3-8b-instruct and mistral/gemma-7b-instruct as Roleplayers, and GPT-4 as Oracle.	Win rate of responses generated by the fine-tuned Questioner against the initial model's responses (human/model preference judgments); log-probability of gold responses as the training reward signal.	After two iterations of self-improvement, the Questioner asks better questions, generating responses preferred over the initial model's on 72% of tasks. Using a strong Questioner/Roleplayer pairing (llama3-8b-instruct with GPT-4 as Oracle) yielded a 66% win rate, suggesting STaR-GATE might work without a significantly more powerful oracle if the Questioner itself is strong enough. Win rates dropped when replacing the training Roleplayer (mixtral) with weaker Roleplayers (mistral, gemma) at test time.	Explicit 'Limitations and Future Work' section: (1) dependence on gold responses (labels) generated by an external Oracle - cannot yet be framed as full self-play/self-improvement, though exploratory self-oracle results are promising; (2) observed drop in win rates when the test-time Roleplayer differs from the training Roleplayer, highlighting the need for training with multiple Roleplayers for robustness (left as future work); (3) did not ablate dataset size, which likely has a strong impact on performance; (4) did not evaluate the fine-tuned model on other domains beyond the training setting; (5) multiple iterations of finetuning are computationally more intensive than a single run, and reducing this cost is left to future work.	Assumes a synthetic persona-task-Roleplayer simulation is a valid proxy for real user preference elicitation. Assumes an external Oracle model with access to the 'true' persona can generate valid gold responses to use as a reward signal - i.e., that ground-truth optimal responses are definable and obtainable at training time. Assumes the training-time Roleplayer's behavior generalizes to different Roleplayers/users at test time (an assumption the paper's own ablation shows is imperfect).	No real human user studies (Roleplayer is always a simulated LM persona, not a human). No voice/speech modality. Alternative RL optimization methods (e.g. REINFORCE with variance reduction) are suggested but not tried, left to future work. Testing on domains beyond the synthetic persona-task setting used for training (explicitly named as an important extension for future work).
18	3	Learning to Clarify: Multi-turn Conversations with Action-Based Contrastive Self-Training (ACT)	Chen et al	2025	https://arxiv.org/abs/2406.00222	LLM-based conversational agents lack conversational skills such as disambiguation - faced with ambiguity, they often overhedge (generate overly-qualified answers) or implicitly guess the user's true intent rather than asking a clarifying question. High-quality task-specific conversation samples are often limited, bottlenecking dialogue-policy learning.	Proposes Action-Based Contrastive Self-Training (ACT): a quasi-online preference-optimization algorithm based on Direct Preference Optimization (DPO), enabling data-efficient dialogue policy learning even when no explicit action label is available, by contrasting an agent's possible pragmatic conversational actions (e.g. clarify vs. answer directly vs. overhedge).	Multiple real-world conversational tasks: tabular-grounded question-answering (based on Deng et al. 2022), machine reading comprehension, and AmbigSQL, a novel task introduced in this paper for disambiguating information-seeking requests for complex SQL generation toward data-analysis agents. Corpora used include PACIFIC (high inter-annotator agreement, 0.62) and Abg-CoQA (lower agreement, 0.26).	Conversation-modeling improvement over standard tuning approaches (supervised fine-tuning, standard DPO); implicit action-recognition evaluation assessing whether the model's chosen dialogue action matches the benchmark's labeled 'optimal' action.	ACT demonstrates substantial conversation-modeling improvements over standard SFT and DPO tuning approaches in data-efficient regimes, across all three tested tasks. Even with a greatly increased number of in-context examples (50-250) for a frontier model via many-shot in-context learning, disambiguation ability does not improve uniformly - a smaller model tuned with ACT and the same conversation samples can outperform frontier models using many-shot in-context examples.	Explicit Appendix section 'A.1 Discussion of Limitations': relies on the assumption that clarification questions in training data are appropriately timed, but crowdsourced conversation datasets are often noisy, which can result in suboptimal learned policies; label noise affects the implicit action-recognition evaluation, more so for low-agreement corpora like Abg-CoQA (0.26) than high-agreement ones like PACIFIC (0.62); ACT uses task-specific heuristics requiring customization/engineering effort; the overall tuning/evaluation pipeline makes heavy use of existing commercial LLMs (Gemini) for tasks like Action Classification or User Simulation, which are imperfect and may not be accessible to all researchers for financial/privacy reasons; the study focuses specifically on the limited-data regime.	Assumes conversational actions (e.g. clarify, answer, overhedge) can be meaningfully discretized and contrasted via a preference-optimization framework. Assumes benchmark-labeled 'optimal' actions in existing datasets are a valid ground truth despite noted annotator disagreement in some corpora. Assumes LLM-based (Gemini) action classification and user simulation are reliable enough substitutes for human annotation/interaction during training and evaluation.	Real human user studies (evaluation relies on benchmark-labeled optimal actions and LLM-based simulation, not live human interaction). No voice/speech modality. Scenarios beyond the limited-data regime the study focuses on. Generalization of ACT-trained policies to conversational domains not covered by the three evaluated tasks.
19	3	Modeling Future Conversation Turns to Teach LLMs to Ask Clarifying Questions	Zhang, Knox & Choi	2025	https://arxiv.org/abs/2410.13788	LLMs often respond to ambiguous user requests by presupposing a single interpretation rather than asking a clarifying question, frustrating users who intended a different interpretation. The authors speculate this stems from standard preference-data labeling practice, where annotators rank responses based only on prior context (single-turn), making it hard to assess the utility of a clarifying question versus a complete-but-presumptuous answer.	Proposes double-turn preference annotation: instead of ranking only the next single assistant turn, annotators interact with the LLM by providing responses to its clarifying questions and observing the LLM's subsequent response, then assign preferences based on whether the completed two-turn interaction successfully fulfilled the request. Trains systems using these double-turn preferences via standard preference-optimization pipelines and compares against training with standard single-turn preferences.	Open-domain QA datasets with multiple annotations per question, derived from Natural Questions (Kwiatkowski et al. 2019) and AmbigQA (Min et al. 2020); evaluated across three base LLMs (referencing Dubey et al. 2024 / Llama and Mesnard et al. 2024 / Gemma family models).	F1 measured against the answer set from different interpretations of each query (effectiveness); number of model turns (efficiency); accuracy of judgments on whether clarification is needed at all.	Training with double-turn preferences achieves a 5% improvement in F1 over training with standard single-turn preference labels, measured against the answer set from different user interpretations, consistent across three base LLMs. Double-turn preferences can also be used to train models to judiciously determine when to ask clarifying questions versus answering directly, achieving a 3% accuracy improvement in this judgment over existing methods.	Explicit 'Limitations and Future Work' section: only considers systems for one- or two-turn interactions, not cases where additional clarifying questions might be helpful after observing the user's first clarifying answer; does not model dialogue acts beyond predicting a single answer or asking one clarifying question (e.g. no Overton responses covering multiple interpretations, no long-vs-short answer selection, no abstaining in lieu of an erroneous answer) - these are named as directions for future work; extending the double-turn method to other tasks is noted as requiring (1) multiple annotator labels for expected outputs and (2) a changeable end-task evaluation metric.	Assumes annotator preference data collected via a double-turn (simulate-then-judge) protocol is a valid, generalizable signal for training - a more expensive but presumably more accurate labeling process than standard single-turn ranking. Assumes the underlying cause of poor clarification behavior is specifically the single-turn nature of standard preference-data collection (a causal claim the paper's improved results are offered as evidence for, though other causes are not ruled out).	General multi-turn interactions beyond one or two turns (explicitly flagged as future work). Other dialogue acts beyond binary answer-vs-clarify (Overton responses, long/short answer selection, abstention) - explicitly flagged as future work. No voice/speech modality. Extension of the double-turn preference method to tasks beyond open-domain QA (framed as a future direction requiring new annotation and metrics).
20	3	CLAMBER: A Benchmark of Identifying and Clarifying Ambiguous Information Needs in Large Language Models	Zhang et al.	2024	https://arxiv.org/abs/2405.12063	The effectiveness of LLMs in dealing with ambiguous user queries remains unknown despite their increasing use to meet information needs, risking user trust and satisfaction. Prior investigations of LLMs and ambiguity are fragmented, lacking a comprehensive taxonomy and often limited to contextual ambiguity, ignoring LLM-specific ambiguity from conflicting internal knowledge.	Introduces a taxonomy of query ambiguity combining three primary dimensions (drawing on the input-process-output framework for evaluating collaborative systems) into eight fine-grained categories. Builds ~12K data items using this taxonomy (the CLAMBER benchmark) and evaluates off-the-shelf LLMs' ability to identify ambiguity and generate clarifying questions, including with chain-of-thought (CoT) and few-shot prompting as baselines.	CLAMBER: ~12K high-quality data items across eight ambiguity categories, evaluated using 5 off-the-shelf LLMs including ChatGPT, constrained by computational resources.	Ambiguity-identification accuracy; clarifying-question quality (assessed via the benchmark's evaluation protocol, details not fully extracted in pages read).	ChatGPT outperforms other smaller-scale LLMs, especially in identifying/clarifying multifaceted queries (e.g. 'What is the largest manufacturer in China?' - ambiguous on the type of manufacturer). Despite CoT and few-shot prompting, current LLMs struggle to identify ambiguities; these techniques may cause overconfidence in small-scale LLMs, negatively impacting ambiguity identification, with only marginal improvement even with many shots and CoT. LLMs struggle to leverage contextual cues to disambiguate pronouns. Current LLMs fail to ask high-quality clarifying questions due to inability to recognize their own knowledge gap - even when they recognize a query is ambiguous, lack of conflict resolution and inaccurate use of inherent knowledge causes uncertainty about which ambiguity to clarify.	Explicit 'Limitations' section: (1) Sensitivity of Prompts - results are likely sensitive to prompt wording; while three different prompts were used and averaged, it's hard to assert they are optimal for the specific issue, and prompt sensitivity/optimality is flagged as a significant open research area; (2) Limited LLMs - only 5 LLMs evaluated due to computational constraints; evaluating additional models (e.g. PaLM 540B) would be advantageous with more resources.	Assumes the eight-category taxonomy (derived from the input-process-output framework) comprehensively captures the space of query ambiguity types relevant to LLMs, including LLM-specific ambiguity from conflicting internal knowledge (not just contextual/external ambiguity as in prior taxonomies). Assumes CoT and few-shot prompting are representative baselines for testing whether prompting alone can fix ambiguity-handling deficits.	Only 5 LLMs evaluated (explicitly named as a computational-resource limitation, not a design choice). No exploration of methods beyond prompting (e.g. no finetuning-based fix attempted for the identified failure modes). No voice/speech modality. No end-to-end interactive dialogue evaluation with real users following a clarifying question.
21	3	ClarQ-LLM: A Benchmark for Models Clarifying and Requesting Information in Task-Oriented Dialog	Gan, Li, Xie, Wen, Purver & Poesio	2024	https://arxiv.org/abs/2409.06097	Task-oriented dialogue often requires an information-seeker agent to ask clarification questions to gather required details from a provider, due to task complexity, the seeker's lack of crucial details, and/or ambiguity or incompleteness in the provider's input. Existing benchmarks (e.g. ShARC and variants) are limited to yes/no clarifying answers with knowledge points listed in advance, making the task appear simpler than realistic dialogue, and rely on surface-form metrics like BLEU rather than semantic/intent correctness.	Introduces ClarQ-LLM: an evaluation framework with bilingual (English-Chinese) conversation tasks, seeker and provider conversational agents, and evaluation metrics. Unlike fixed-content benchmarks, includes a provider conversational agent (powered by GPT-4o or LLAMA3.1-405B) that replicates the original human provider, allowing seeker agents to be tested by directly interacting with the provider agent rather than against static dialogue content. Success is measured by whether the seeker agent gathers all necessary information, regardless of the specific order or phrasing of questions asked.	31 different task types, each with 10 unique dialogue scenarios between information seeker and provider agents (e.g. restaurant ordering), spanning English and Chinese.	Task success rate (whether the seeker agent gathered all necessary information from the provider agent), contrasted explicitly against traditional fluency/match-based metrics like BLEU and dialogue-state-tracking accuracy, which the authors argue are inadequate for this setting.	Even state-of-the-art models struggle: LLAMA3.1 405B as seeker agent achieved a maximum success rate of only 60.05%, and GPT-4o achieved 50.8%, both significantly lower than the 85% success rate achieved by human seekers. A response that correctly guesses the customer's needs without asking clarification is still scored as incorrect, since ClarQ-LLM considers reliance on guesswork - even if occasionally correct - unable to ensure consistently accurate responses.	No dedicated 'Limitations' section was found in the pages examined (paper has an Ethical Considerations section instead, addressing content safety choices such as avoiding tasks that could harm people/animals, and fair payment for student annotators). One methodological limitation noted in-text: the provider agent originally could only give one piece of information to the seeker at a time; because some seeker agents (in 'Completion' mode) ask multiple questions simultaneously, the authors modified the provider agent to combine multiple responses into a single reply (the 'multi-info provider agent') to prevent this provider limitation from unfairly affecting seeker agent performance.	Assumes task success (gathering all necessary information) is the appropriate ground truth for evaluating a seeker's clarification ability, rather than matching a specific human dialogue trajectory - a deliberate departure from BLEU/state-tracking-style evaluation. Assumes a GPT-4o- or LLAMA3.1-405B-powered provider agent can faithfully replicate the genuine needs of the original human provider across arbitrary question orderings.	No voice/speech modality (text-based task-oriented dialogue). No evaluation of the provider agent itself - the benchmark is explicitly designed to evaluate only the seeker agent, not the provider's realism or accuracy as a simulated human. No discussion of confidence/calibration signals driving when the seeker decides to ask versus guess.

22	4	The Theory and Use of Clarification Requests in Dialogue	Purver (PhD Thesis, King's College London)	2004	https://www.researchgate.net/publication/2897205_The_Theory_and_Use_of_Clarification_Requests_in_Dialogue_Matthew_Purver	Clarification requests (CRs) are an important, relatively common, and yet under-studied dialogue device allowing a user to ask about some feature (e.g. meaning or form) of an utterance or part thereof; the thesis aims to understand how CRs arise, what forms they take, what they mean, how they are responded to, and how this understanding can add clarification capabilities to computational dialogue systems.	Combines empirical, theoretical, and implementational work: (1) corpus-based and experimental studies establishing a taxonomy of CR types, sources, forms and readings; (2) a syntactic/semantic analysis using the HPSG framework, extending Ginzburg and Cooper (2004), including a witness-set-based representation for noun phrases; (3) a prototype text-based dialogue system, CLARIE, implemented in Prolog using the TrindiKit, combining GoDiS's information-state-based dialogue management with SHARDS's HPSG-based ellipsis resolution, adding the capability to generate and interpret CRs for incomprehensible/contradictory input, unknown/ambiguous reference, and unknown vocabulary.	British National Corpus (BNC): a 10-million-word sub-corpus of English dialogue transcripts, of which a sub-portion of dialogue transcripts was used and manually marked up (multi-layered annotation following BNC SGML conventions) for the corpus investigations into CR ontology, sources, and responses.	Not primarily quantitative benchmark metrics (this is a linguistic/theoretical thesis); empirical chapters report corpus frequency/distribution statistics and experimental results on CR interpretation and generation, evaluated via the developed taxonomy rather than standard NLP metrics.	CRs are a relatively common phenomenon in dialogue; function words are rarely sources of CRs (rarely clarified, and echoed function words are hard to interpret as reprise fragments), while nouns and noun phrases are commonly clarified; verbs are also rarely CR sources despite experiments suggesting echoed verbs could be interpreted as reprises just as readily as nouns, suggesting something about verb semantics/information-content/mutual-knowledge makes them less likely targets. Many CRs are not answered, but those that are tend to be answered in the very next turn (unlike other question types in general, though similar to yes/no questions). CRs behave more like standard questions (introducing questions under discussion) than as a special immediate-repair grounding act, since other Q&A pairs can intervene before a CR is answered. Reprises of NPs concern individual referents while reprises of daughter determiners/nouns do not, implying a grammar should not assume NPs inherit content directly from daughters.	Explicit Chapter 7 'Arising Issues and Further Work': the CLARIE implementation is only a prototype, leaving much to be desired as a practical usable system (limited lexicon, limited-coverage grammar, unrealistic domain, no speech interface); adding speech recognition is flagged as challenging, particularly the interaction of a standard recognizer (which makes best guesses at known words) with the thesis's treatment of unknown words, though a dual-recognizer approach is suggested as promising; little attention was paid to disambiguating fragments between CR and non-CR readings, or to identifying a CR's source when more than one potential source exists in the most recent utterance - author suggests a probabilistic/machine-learning approach may eventually be needed as more information sources (antecedent syntax, semantics, intonation) are incorporated beyond the simple ordered grounding rules used.	Assumes CR generation/interpretation can be achieved via a simple, non-heavyweight grounding process operating on contextually-dependent grammatical representations, without having to model or reason explicitly about the other participant's beliefs. Assumes the same grounding process applies symmetrically to both system-generated and user-generated CRs. The thesis's speech-related future-work discussion explicitly flags that confidence scores (citing Gabsdil and Bos 2003) could potentially be combined with the CR approach to allow explicit/implicit confirmation behavior - i.e., assumes confidence-driven CR triggering is a promising but unexplored extension, not something the thesis itself implements.	Speech/spoken input - the CLARIE system and empirical work are entirely text-based; using low ASR confidence scores to prompt lexical CRs is explicitly named as a future-work idea (citing Larsson 2002), not something tested in the thesis. Intonation/prosody as a disambiguation signal for CR forms and readings - flagged as a promising future direction given evidence reprise questions can be distinguished from statements via pitch contour, but not implemented. A probabilistic/machine-learning approach to CR source/type disambiguation when multiple potential sources exist - explicitly flagged as likely necessary but not attempted.
23	4	POMDP-Based Statistical Spoken Dialog Systems: A Review	Young et al.	2013	https://www.researchgate.net/publication/260686005_POMDP-Based_Statistical_Spoken_Dialog_Systems_A_Review	Conventional flowchart-based spoken dialog systems (SDSs) are expensive to build and fragile in operation, especially given persistent speech-recognition word error rates of 15-30% in real-world noisy environments; a data-driven framework is needed that reduces handcrafting costs and provides robustness against recognition errors.	Reviews the POMDP (partially observable Markov decision process) framework for SDSs: dialogue is modeled as a Markov process with a hidden state, updated via Bayesian belief monitoring over a distribution of possible states rather than a single most-likely state; the system's output (SLU) is treated as a noisy observation of the true user intent; policy (the mapping from belief state to system action) is optimized via reinforcement learning against a reward function, either offline from dialogue corpora or online through interaction with real users.	Not a single dataset - this is a review covering the POMDP-SDS literature broadly, including reference to the CMU Let's Go bus-information system used in the Spoken Dialog System Challenge (a live public service in Pittsburgh, PA) as a real-world evaluation testbed.	Reviews metrics used across the POMDP-SDS literature, notably dialogue success rate as a function of word error rate (WER), via logistic regression of success against WER for systems tested in the 2010 Spoken Dialog Challenge.	The belief state's explicit representation of uncertainty leads to systems more robust to speech recognition errors, since evidence is integrated across turns so a single error has reduced impact and user persistence is rewarded. Pursuing all possible dialogue paths in parallel (via the belief distribution) removes the need for explicit backtracking or error-correction dialogs. In the 2010 Spoken Dialog Challenge live test, a hand-coded baseline system was slightly better at low error rates (reflecting its long development/refinement history), but two POMDP-based systems, despite much shorter development times, were more robust than the hand-coded baseline at high noise levels.	Explicit 'Conclusion' section: exact POMDP model representation and optimization is computationally intractable, requiring efficient algorithms and carefully constructed approximations for practical use; POMDP-based systems remain complex and involve approximations and tradeoffs, with much still to do - specifically named challenges include increasing dialogue-model complexity while maintaining tractable belief tracking, reducing policy-learning times so systems can train directly on real users rather than simulators, and packaging the technology to make it accessible to non-expert industrial practitioners. The framework depends critically on reward signals, but it is very difficult in practice to extract reliable reward signals from users, even for a simple success/fail criterion.	Assumes dialogue can be validly modeled as a Markov process (state transitions depend only on the current state and action, not full history). Assumes the SLU output can be treated as a single noisy observation with a well-defined observation probability function, and that transition/observation probabilities can be represented by a tractable stochastic model. Assumes a reward function can be meaningfully specified to encode desired dialogue-system behavior, despite the paper's own acknowledgment that reliable reward signals are hard to extract from real users.	Multimodal dialog (gestures, visual displays, haptic feedback) is explicitly noted as beyond the paper's scope, though the POMDP framework is noted as extensible to it, at the cost of larger state spaces and synchronization issues. The review itself conducts no new experiments; it does not test or propose specific solutions to the noted intractability of increasing model complexity while maintaining tractable belief tracking - this is named as an open challenge, not solved.
24	4	Robust Speech Recognition via Large-Scale Weak Supervision (Whisper)	Radford, Kim, Xu, Brockman, McLeavey & Sutskever (OpenAI)	2022	https://arxiv.org/abs/2212.04356	Unsupervised pre-trained audio encoders (e.g. Wav2Vec 2.0) learn high-quality speech representations but lack an equivalently performant decoder, necessitating a fine-tuning stage to perform tasks like speech recognition; fine-tuning is a complex process requiring skilled practitioners and risks the model learning brittle, dataset-specific patterns that don't generalize (a documented risk in computer vision transfers analogously here). The goal is a speech recognition system that works reliably 'out of the box' across a broad range of environments without dataset-specific fine-tuning.	Trains a sequence-to-sequence Transformer ('Whisper') simply to predict large amounts of raw internet audio transcripts, scaled to 680,000 hours of multilingual and multitask supervision (117,000 hours covering 96 non-English languages; 125,000 hours of X-to-English translation data), taking a minimalist data-preprocessing approach that predicts raw transcript text without significant standardization. Constructs the dataset via automated filtering heuristics to detect and remove likely machine-generated transcripts, an audio-language-detector to match spoken and transcript language, and fuzzy de-duplication.	680,000 hours of audio-transcript pairs scraped from the internet; broken into 30-second segments (including some segments with no speech, sub-sampled, for voice-activity-detection training). Evaluated zero-shot (without any dataset-specific fine-tuning) against standard speech-recognition benchmarks.	Word error rate (WER) and related zero-shot transfer performance metrics across standard ASR benchmarks, compared against prior fully-supervised results and against human transcription accuracy/robustness.	Models trained at this scale generalize well to standard benchmarks and are often competitive with prior fully-supervised results in a zero-shot transfer setting, without any dataset-specific fine-tuning. When compared to humans, Whisper models approach human-level accuracy and robustness. This suggests simple scaling of weakly-supervised pre-training was underappreciated for speech recognition prior to this work, achieved without needing the self-supervision/self-training techniques common in recent large-scale ASR work.	Explicit 'Limitations and Future Work' section: remaining errors in long-form transcription are non-perceptual failure modes of seq2seq models (getting stuck in repeat loops, missing first/last words of a segment, complete hallucination of unrelated transcript text) - decoding-detail mitigations help but the authors suspect further fine-tuning or RL-based decoding optimization could help more; performance is still quite poor on many lower-resource languages, since the pre-training dataset is heavily English-biased from the data-collection pipeline sourcing primarily English-centric internet content, with most languages having less than 1000 hours of training data; the paper studies only zero-shot transfer, not fine-tuning, though fine-tuning is likely to improve results further for domains with high-quality supervised data available; it's currently unclear whether Whisper's robustness stems from its encoder, decoder, or both, since it lacks unsupervised pre-training/self-teaching methods used by other state-of-the-art systems.	Assumes that scale and diversity of weakly-supervised (noisy, internet-sourced) transcripts can substitute for the self-supervised pre-training + fine-tuning paradigm used by prior audio encoders. Assumes automated heuristic filtering (removing likely machine-generated transcripts, language-matching, de-duplication) is sufficient to produce a usably high-quality training signal from inherently noisy internet data. Assumes zero-shot transfer performance is the most representative measure of 'general reliability' for a speech recognition system, deliberately not optimizing for any single benchmark via fine-tuning.	Fine-tuning on high-quality supervised speech data for specific domains (explicitly flagged as likely to improve results further, but not studied here since the paper focuses on zero-shot robustness). Ablation isolating whether robustness stems from the encoder, decoder, or both (flagged as unclear and suggested for future study). Auxiliary unsupervised pre-training or self-teaching objectives combined with Whisper's supervised approach (not attempted, though not ruled out as potentially helpful). Targeted data-collection efforts for lower-resource languages (identified as a clear route to improvement but not carried out in this work).
25	4	Moshi: a speech-text foundation model for real-time dialogue	Défossez et al.	2024	https://arxiv.org/abs/2410.00037	Current spoken dialogue systems rely on pipelines of independent components (voice activity detection, ASR, textual dialogue, TTS); this complexity induces multi-second latency between interactions, loses non-linguistic information (emotion, non-speech sounds) since text is the intermediate modality, and relies on turn segmentation that cannot handle overlapping speech, interruptions, or interjections.	Introduces Moshi, casting spoken dialogue as speech-to-speech generation: starting from a text LLM backbone (Helium, 7B parameters), generates speech as tokens from the residual quantizer of a neural audio codec (Mimi), modeling the system's own speech and the user's speech as two parallel autoregressive token streams (removing the concept of explicit speaker turns, allowing arbitrary conversational dynamics including overlap/interruption). Extends hierarchical semantic-to-acoustic token generation with 'Inner Monologue': predicting time-aligned text tokens as a prefix to audio tokens, improving linguistic quality and enabling streaming ASR/TTS as a byproduct. speech → text → LLM → text → speech	- Text dataset (2.1T token) - CommonCrawl (~87% of text data) - Wikipedia, Wikibooks, Wikisource, Wikinews, StackExchange, peS2o - Unsupervised audio dataset (7 million hours) -Fisher (Multi-speaker 2,000 hours) - Supervised multi-stream dataset (170 hours) - Synthetic speech/instruction data (20k+ hours)	The paper evaluates Moshi along four broad dimensions: **1- Text/language understanding:** primarily text/language benchmarks **2- Speech intelligibility:** WER/intelligibility, ABX phonetic discriminability **3- Audio quality:** objective & subjective audio quality **4- Spoken question answering:** spoken-QA performance	Moshi is presented as the first real-time full-duplex spoken large language model, with theoretical latency (160ms) lower than the average human conversational response time (230ms across 10 languages, per Stivers et al. 2009) and demonstrates state-of-the-art spoken question answering and dialogue modeling while displaying satisfactory safety levels (not generating toxic content, remaining voice-consistent) and following complex multi-turn conversations across 5-minute sessions.	No single dedicated 'Limitations' section was located in the extracted pages; limitations are instead discussed contextually - e.g., the semantic tokens produced by the Mimi codec have poor phonetic discriminability (ABX) in the absence of distillation, comparable to prior acoustic-only tokenizers, meaning raw semantic tokens alone are not amenable to capturing linguistic content without distillation; the paper notes a conflict/trade-off between acoustic reconstruction quality and semantic distillation performance (improving one degrades the other, partially mitigated via a split-RVQ design).	Assumes discrete audio tokens from a neural codec (Mimi) are an adequate representation for a text-LLM-derived backbone to jointly reason over speech and text. Assumes modeling the user's and system's speech as two separate parallel streams (rather than a single turn-segmented sequence) is sufficient to capture natural conversational dynamics like interruption and overlap, without needing an explicit turn-taking or interruption-detection module.	No explicit treatment of confidence/uncertainty in the model's own responses, nor any clarification-question behavior - Moshi is designed to always generate a spoken response, not to detect ambiguity or ask the user for more information. No discussion of ASR-confidence-style uncertainty signals despite Moshi effectively performing streaming ASR as a byproduct of Inner Monologue. No evaluation of the system's behavior under genuinely ambiguous or underspecified user requests (evaluation is on spoken QA/dialogue modeling and safety/toxicity, not disambiguation).
26	4	VoiceBench: Benchmarking LLM-Based Voice Assistants	Chen et al	2024	https://arxiv.org/abs/2410.17196	No standardized benchmark exists for evaluating LLM-based voice assistants under realistic conditions; prior evaluations focus primarily on ASR accuracy or general-knowledge QA using clean synthetic speech, disconnected from real-world scenarios involving diverse speaker characteristics, environmental noise, and content variation (grammar errors, mispronunciations, disfluencies).	Introduces VoiceBench, a multi-faceted benchmark using both real and synthetic spoken instructions to evaluate general knowledge, instruction-following ability, and safety, while systematically varying three real-world factors: speaker characteristics (age, accent, pitch, via TTS and voice cloning), environmental conditions (signal distortion, echo, far-field effects), and content variation (grammar errors, mispronunciations, disfluencies, synthesized via state-of-the-art LLMs).	A combination of real and synthetic spoken instruction data spanning general knowledge, instruction-following, and safety test cases, augmented with speaker/environment/content perturbations layered on top.	Task performance on general knowledge, instruction-following, and safety measurement dimensions; robustness comparison across the speaker/environmental/content perturbation conditions.	Reveals a significant performance gap between end-to-end voice assistant models and traditional pipeline models that combine a separate ASR system with an LLM - evident not only in overall performance but also in robustness across the tested speaker/environmental/content variations, indicating current evaluation protocols (over-reliant on ASR-only or synthetic-clean-speech assessment) understate real-world weaknesses that VoiceBench newly surfaces.	No dedicated 'Limitations' section was located in the pages examined, however, the paper's Conclusion frames the identified performance gaps and vulnerabilities as findings pointing to needed future work (underscoring the need for further advancements in end-to-end voice assistant processing of spoken instructions), rather than stating limitations of VoiceBench's own methodology.	Assumes TTS and voice-cloning-generated speech with systematically varied speaker/environmental/content properties is a valid proxy for genuinely diverse real-world spoken input. Assumes general knowledge, instruction-following, and safety are the three key aspects worth benchmarking for voice assistants, by analogy to how text-based LLMs are typically evaluated.	No discussion of clarification-question behavior or ambiguity handling - the benchmark evaluates robustness to acoustic/speaker/content perturbation and general capability, not whether the assistant recognizes when it should ask the user for more information. No confidence/calibration-signal evaluation. Multi-turn dialogue - the described evaluation is centered on general knowledge, instruction-following and safety per spoken instruction, not sustained multi-turn conversation.
27	4	SD-Eval: A Benchmark Dataset for Spoken Dialogue Understanding Beyond Words	Ao et al.	2024	https://arxiv.org/abs/2406.13340	Speech carries content, paralinguistic (who says/how said - accent, age, timbre, prosody, volume, rhythm), and environmental information, all of which significantly impact communication; chat-oriented LLMs extended to handle speech input can often recognize/analyze speech but fall short of generating appropriate responses, due to a lack of principled task definitions and suitable open-source datasets/metrics for evaluating this beyond-words dimension.	Introduces SD-Eval, a benchmark dataset for multidimensional evaluation of spoken dialogue understanding and generation, focused on paralinguistic and environmental information. Aggregates data from eight public datasets representing four perspectives (emotion, accent, age, background sound) into four sub-tasks (test-emo, test-acc, test-age, test-env). Implements three different models and constructs a matching training set to evaluate the benchmark, using objective metrics (BLEU, ROUGE), subjective human evaluation, and LLM-based metrics for generated responses.	7,303 utterances (8.76 hours of speech), aggregated from eight public datasets across the four sub-tasks (emotion, accent, age, background sound); training set of 1,052.72 hours of speech data / 724.4k utterances constructed via a similar process to SD-Eval itself.	Objective evaluation methods (BLEU, ROUGE), subjective opinion-score human evaluation, and LLM-based metrics for the generated spoken-dialogue responses.	Models conditioned with paralinguistic and environmental information outperform their counterparts (lacking this conditioning) in both objective and subjective measures. LLM-based metrics show a higher correlation with human evaluation compared to traditional metrics like BLEU and ROUGE for this task.	Explicit 'Limitations and Future Work' section: SD-Eval accommodates only speech-to-text dialogues, limiting evaluation of system responses at the text level (not speech-to-speech); currently supports only single-turn dialogue evaluation, limiting application to more complex multi-turn interactions; the four sub-tasks cover emotion, accent, age, and environmental information but do not yet account for other aspects such as speaker gender - addressing these gaps, with the ultimate goal of a benchmark for multi-turn speech-to-speech dialogue evaluation, is named as future work.	Assumes that aggregating and relabeling eight distinct public datasets (originally collected for different purposes) into a unified four-perspective framework produces a valid, coherent benchmark for paralinguistic/environmental understanding. Assumes objective metrics like BLEU/ROUGE, subjective human scores, and LLM-based judges are each meaningful (if imperfect) proxies for response appropriateness, motivating the comparison of correlation across all three.	Speech-to-speech dialogue (only speech-to-text is evaluated, explicitly named as a limitation and future goal). Multi-turn dialogue (only single-turn is currently supported). Speaker gender as a paralinguistic dimension (not included in the current four sub-tasks, named as future work). No treatment of ambiguity, clarification, or confidence/calibration - the benchmark's focus is response appropriateness given non-verbal cues, not disambiguation behavior.
28	4	Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction	Gosai, Vuong, Tyagi, Li, You, Bavare, Ucar, Fang, Jang, Liu & He (Scale AI)	2025	https://arxiv.org/abs/2512.14865	End-to-end (E2E) spoken dialogue systems are increasingly replacing cascaded ASR+LLM+TTS pipelines, but existing benchmarks primarily evaluate them on synthetic speech and single-turn tasks, leaving realistic multi-turn conversational ability (e.g. tracking edits, audio cues, long-range context) underexplored; strong single-turn performance in text-only LLMs is known not to guarantee robust multi-turn behavior, and this weakness is amplified in the audio modality since audio is temporally continuous and high-frame-rate, rapidly consuming context.	Introduces Audio MultiChallenge, extending the text-based MultiChallenge framework (which evaluates Inference Memory, Instruction Retention, and Self Coherence) to audio, adding a new axis, Voice Editing, testing robustness to mid-utterance speech repairs/backtracking, and augmenting existing axes with audio-native elements (e.g. Audio-Cue Inference Memory requiring recall of ambient sounds/paralinguistic signals, not just semantic content). Data generated via a hybrid audio-native agentic + human-in-the-loop pipeline: human contributors are given high-level blueprints/interaction goals (not verbatim scripts) derived from a scalable agentic loop designed to elicit diverse, realistic failure modes while preserving natural disfluency. Evaluation uses 1,712 atomic, instance-specific rubrics per conversation (LLM-as-judge) rather than coarse binary success metrics.	452 multi-turn conversations from 47 unique speakers, totaling 15 hours of audio, with 1,712 instance-specific rubrics decomposing each ideal response into critical sub-requirements.	Rubric-based pass rate (a task passes only if all corresponding rubrics are satisfied); the LLM-as-a-judge rubric method achieves 93% agreement with human evaluators, validating it as a proxy metric.	Even frontier models struggle on the benchmark: the highest-performing model tested (Gemini 3 Pro Preview, Thinking mode) achieves only a 54.65% pass rate. Error analysis shows models fail most often on the newly introduced axes (Voice Editing, Audio-Cue Memory), and Self Coherence specifically degrades as audio context length increases - reflecting difficulty tracking edits, audio cues, and long-range context in natural spoken dialogue.	Explicit statement in the 'Conclusion and Future Work' section: 'A key limitation of this work is that our evaluation does not operate directly on raw audio outputs.' Future work is suggested to explore improved hybrid data-curation pipelines, longer conversational contexts, and open-ended audio-native evaluation frameworks, which could enable more principled assessment of paralinguistic instruction retention and surface open questions around rubric design, verification, and the role of audio-native judges.	Assumes fine-grained, instance-specific rubrics (rather than coarse pass/fail judgments) are necessary and sufficient to meaningfully attribute model failures in multi-turn spoken dialogue. Assumes an LLM-as-judge, validated at 93% agreement with human evaluators, is an adequate substitute for full human evaluation at scale. Assumes the four axes (Inference Memory, Instruction Retention, Self Coherence, Voice Editing) comprehensively cover the key interaction patterns needed for natural multi-turn spoken dialogue.	Evaluation directly on raw audio model outputs (explicitly named as the paper's own key limitation - the evaluation pipeline does not operate on raw audio). Longer conversational contexts beyond what's included in the current 452-conversation, 15-hour dataset (flagged as future work). Open-ended (non-rubric-based) audio-native evaluation frameworks (flagged as future work). No treatment of confidence/calibration signals or explicit clarification-question behavior as a distinct evaluated axis (the closest related axis, Voice Editing, evaluates robustness to the user's self-corrections, not the system's own decision to ask for clarification).
29	4	Sorry, I didn't catch that! - An investigation of non-understanding errors and recovery strategies	Bohus & Rudnicky	2005	https://aclanthology.org/2005.sigdial-1.14/	Spoken dialogue systems (SDS) frequently suffer from ASR recognition errors and non-understandings. How can dialogue managers detect non-understanding errors and dynamically trigger appropriate recovery/confirmation strategies (explicit vs. implicit confirmation, ask-repeat, yield)?	Evaluates error recovery strategies in a real-world task-oriented spoken dialogue system (RavenClaw architecture). Uses heuristic and data-driven rules based on ASR confidence scores, signal cues, and dialogue state to switch between explicit confirmation ("Did you say X?"), implicit confirmation, and re-prompting.	Task-oriented spoken dialogue system (CMU Let's Go / Communicator telephone bus information domain, spoken interactions with human callers).	Task completion rate, turn efficiency, error recovery success rate, user satisfaction, dialogue length.	Explicit confirmation is highly effective when confidence is low-to-medium, whereas implicit confirmation works best when confidence is high. ASR confidence thresholds can effectively govern error recovery strategy selection.	Restricted to rigid, domain-specific slot-filling state machines (e.g., bus routes/times); strategies are hand-coded/rule-based rather than learned end-to-end; limited to finite-state slot grammars.	Assumes a pre-defined slot-filling ontology; assumes ASR confidence score is available and correlated with recognition accuracy; assumes fixed set of dialog actions.	Open-domain LLMs or generative free-text responses. No notion of Value of Information (EVPI) or soft cognitive cost optimization. No joint modeling of acoustic SNR and generative semantic entropy.
30	4	Partially observable Markov decision processes for spoken dialog systems	Williams and Young	2007	https://www.sciencedirect.com/science/article/abs/pii/S0885230806000283?via%3Dihub	Traditional spoken dialogue managers track conversation state using deterministic heuristics, which fail under noisy ASR conditions. How can dialogue managers maintain uncertainty over user intent and optimize dialogue actions under observation noise?	Formulates spoken dialogue management as a Partially Observable Markov Decision Process (POMDP). Represents user intent as a hidden state, maintains a belief distribution b(s) updated via noisy ASR observations, and uses Reinforcement Learning / point-based value iteration to learn an optimal policy balancing action reward against risk of misinterpretation.	Simulated and real spoken dialogue interactions (e.g. Tourist Information System, dialogue domains with noisy speech channels).	Expected reward, task success rate, average dialogue length, reward vs. WER (Word Error Rate) curves.	POMDP-based dialogue policies significantly outperform deterministic finite-state managers in high-noise speech environments. The policy automatically learns to ask for confirmation when belief state entropy is high, and acts directly when belief state is certain.	Scalability ("curse of dimensionality") restricts exact POMDP solving to small, hand-crafted state spaces and slot ontologies. High computational complexity for state belief updating.	Assumes discrete, finite state spaces and hand-designed slot ontologies; assumes known or estimated observation channel probabilities P(o|s), assumes fixed user goal per dialogue.	Open-domain generative large language models (LLMs) with unconstrained natural language intent spaces. No dynamic adjustment of question phrasing types (open vs. closed vs. confirm) based on cognitive burden vs. SNR. No integration of semantic entropy over text generation logprobs.
31	3	Value of Information: A Framework for Human–Agent Communication	Dong et al.	2026	https://arxiv.org/abs/2601.06407	How can an LLM agent decide when to ask a user a clarifying question versus acting directly, balancing the utility of getting a better answer against the communication cost/burden imposed on the user?	Applies Expected Value of Perfect Information (EVPI) from decision theory to LLM agents. Computes expected utility of acting under current belief vs. expected utility after asking a clarification question minus a fixed interaction cost c. Uses Monte Carlo sampling over LLM belief states to estimate information gain.	Text-based ambiguous QA and task-oriented dialogue benchmarks (for example: AmbigQA, ClarQ).	Expected Utility, Task Success Rate, Clarification Precision/Recall, Communication Cost / Turn Count.	EVPI-based decision policies significantly outperform fixed confidence threshold baselines, asking questions selectively only when the expected utility gain exceeds communication cost c. Prevents over-clarification on low-stakes or unambiguous queries.	Assumes a perfect observation channel—meaning if the agent asks a question, the user's answer is received with 100% accuracy (p(^y|y)=1). Ignores physical noise, acoustic speech errors, or communication channel corruption.	Assumes user responses are provided as clean text without transcription noise; assumes fixed communication cost **c** regardless of question type or channel quality; assumes static belief updating.	Any speech/audio modality, ASR recognition errors, or environmental acoustic noise (SNR). Does not test varying question structures (open vs. closed vs. confirm) to adapt to observation channel noise. Does not test multi-turn compounding acoustic noise.